

Graduate Research Plan Statement

It is well established observationally that galaxies in the local Universe follow a strong bimodal color distribution. Galaxies that are star-forming show strong blue colors while galaxies that have stopped forming stars and contain predominantly old stellar populations are red, known as "quiescent" galaxies (QGs). Presently, 100% of the most massive ($\log M/M_{\text{sol}} > 11$) galaxies are quiescent (e.g., forming stars well below the stellar mass star-formation rate relation at a given redshift, z). Modern simulations and theoretical models show that a combination of physical effects (e.g. nuclear black hole activity, violent disk instabilities, mergers, etc.) can recover this present-day quiescent fraction. These same models also predict that the quiescent fraction of massive galaxies declines with increasing redshift, such that massive QGs should be fairly rare at $z > 2$ ($\sim 10\text{-}20\%$)^[1]. This, however, is countered by recent observations at $z > 3$. Past surveys over small fields have found varying quiescent fractions among massive galaxies (e.g. 16%-40%^[2,3]). QGs have been found to even higher redshifts, with several confirmed at $3.5 < z < 4.0$ ^[4]. While differing definitions of "quiescence" and Poisson & cosmic variance (CV) uncertainties dominate these disagreements, *all of these observations are in tension with theoretical predictions*, which find the quiescent fraction to be very low at $z \sim 3$, and zero by $z > 4$. Clearly, even the current most sophisticated models are missing physical processes at high- z that are necessary to rapidly quench star formation during the first ~ 1.5 billion years of the Universe. To solve this mystery, we must now turn to wide-field optical and infrared surveys in order to overcome previous uncertainties and begin to uncover the mechanisms governing these enigmatic sources.

Intellectual Merit In order to contribute to the understanding of the evolution of massive galaxies in the early Universe, **I will robustly quantify the abundance of massive ($\log M/M_{\text{sol}} > 11$) quiescent galaxies at $3 < z < 5$ in the 22 deg² Spitzer-HETDEX Exploratory Large Area Survey (SHELA).** By performing spectral energy distribution (SED) fitting to these galaxies, we can infer their photometric redshifts, star-formation rates, and stellar masses. SHELA includes CTIO/DECam optical *ugrizY* ($m_{\text{5sigma,AB}} \sim 25\text{-}26$), KPNO/NEWMFIRM near-infrared *K* ($m_{\text{5sigma,AB}} \sim 23$; PI Finkelstein), and Spitzer/IRAC mid-infrared ($m_{\text{5sigma,AB}} \sim 22.5$) imaging, probing a 0.4 Gpc³ volume at $z = 3\text{-}5$. By using such a large survey, *I will, for the first time, robustly constrain the quiescent abundance at $z > 3$* , overcoming the sample size and cosmic variance limitations of previous studies. Through spectroscopic follow-up, we can then gain insight into the formation of QGs and understand how galaxies switch from actively star-forming to quenched and what processes and conditions drive this evolution in the early Universe. There are three phases of this program: constructing a redshift luminosity prior and performing SED fitting to candidate massive QGs (year 1), generating a catalog of massive high- z galaxies and robustly constraining the abundance of QGs (years 1-2), and probing massive QGs at even higher redshifts through upcoming *JWST* programs, as well conducting detailed spectroscopic analysis to study the formation histories of these galaxies (year 3).

Phase 1: Due to the red nature of high- z massive QGs, contamination by lower redshift red galaxies is a strong concern. The somewhat limited depth of ground-based optical observations mean that many candidate QGs have SEDs which are also consistent with $z \sim 1$ dusty star-forming galaxies. To reduce contamination from low-redshift galaxies, Stevans et al. (2021; hereafter S21) used a built-in EAZY redshift prior which was generated from the semi-analytic model (SAM) of De Lucia & Blaizot (2007). However, this SAM does not accurately reproduce modern measurements of the $z > 2$ universe^[7] (specifically, red galaxies at $z \sim 3$ are under predicted by a factor of ~ 8). Indeed, S21 found that this prior removed nearly all high- z quiescent candidates, including one confirmed to be at $z = 4.64$. The first phase of this project will focus on improving the selection of these galaxies by building a redshift prior based on modern SAMs more in line with observed galaxy distributions. We chose to study the well established Santa Cruz SAM (SC-SAM)^[8,9], which has been shown to be successful at explaining both local and high-redshift observations. By analyzing the SC-SAM, we can generate a redshift probability prior which will allow us to minimize low redshift contaminants in photometric redshift estimations. **Over the past year, I have completed the preliminary redshift luminosity prior generated from a 2 deg² lightcone populated with galaxies using SC-SAM**, which recovers a factor of ~ 7 more galaxies

compared to the default EAZY prior at $K=22$ at $z\sim 3.5$. To work towards the goal of a robust measure of the quiescent abundance at $z>3$, I will test the validity and calibrate my prior using in-hand ALMA Band 6 observations for 100 candidate massive QGs from S21. Non-detections in ALMA will rule out low- z solutions (which have large fluxes at longer wavelengths due to their dusty natures). At the time of this research plan, I have validated the ALMA Band 6 data and have begun the process of SED fitting.

Phase 2: The completed prior will be applied to an updated SHELA catalog generated by Leung et al. (in prep). This catalog will be >1 mag deeper due to newly obtained DECam imaging (PI Papovich) and will add new photometry in the DECam Y , VICS82 J and K filters. From this catalog, I will perform SED fitting, which will enable me to constrain photometric redshifts, as well as determine the mass and star-formation rates of massive high- z galaxies. **This work will be completed June 2022, and will result in the current largest sample of massive high- z QGs**, from which we will robustly constrain the quiescent abundance at high redshifts as well as allow for comparisons with simulations to constrain the possible mechanisms that are responsible for the formation of these galaxies. This process will be applied not only to SHELA, but to upcoming space based observations such as the COSMOS-Webb program (anticipated mid-late 2023) as well to further push the frontier of discovering these massive QGs.

Phase 3: The COSMOS-Webb Survey (co-PIs Casey & Kartaltepe), is a treasury *JWST* program to map a contiguous 0.6 deg^2 area with deep NIRCcam imaging in 4 filters and a 0.2 deg^2 deep MIRI survey in parallel. By applying my methods to this new dataset, I will be able to probe quiescent galaxies to even higher redshifts of $z\sim 6-7$ (if they exist!) and also study their morphologies across all of $3 < z < 7$. I am also involved in the Cosmic Evolution Early Release Science survey (PI Finkelstein) which covers 100 arcmin^2 with *JWST*, and (in June 2022) is expected to spectroscopically observe ~ 20 massive high- z QGs. We will also utilize data from the *JWST* NIRSpec WIDE MOS Survey in the CANDELS fields (P.I. Ferruit; data public by the last two years of this fellowship), this program will provide spectroscopy of *hundreds* of high- z massive QGs. Spectroscopy will provide crucial redshift confirmations and allow us to constrain the star formation histories of these galaxies. *Larger samples of QGs as well as detailed space-based spectroscopy will allow us to place stringent constraints on the formation of the Universe's most massive galaxies from the Epoch of Reionization to the peak of galaxy assembly.*

I anticipate writing four first author publications from the results of this program, including a paper in progress presenting the redshift luminosity prior. A second will publish the sample of massive galaxies with the quiescent abundance at higher redshifts than even before and comparisons to current simulations. Paper three of this fellowship will publish new samples of massive high- z QGs in the COSMOS-Webb survey, and four will be spectroscopic analysis of *JWST* data from the aforementioned GTOs.

Broader Impacts As a graduate student, I have the unique opportunity of bridging the population between undergrad and young aspiring scientists with researchers established in the field. To help improve diversity and retention in astronomy, I plan to help introduce astronomy as a viable field to young members of underrepresented communities through outreach efforts and involvement in the Outreach for Astronomy in Texas Schools (discussed in detail in personal statement). As I have in the past year at UT, I will continue to mentor both UT Austin students in the Grad/Undergrad Mentorship in Astronomy (GUMMY) program as well as non-UT students through TAURUS, which invites students from traditionally marginalized populations in astronomy to conduct research at UT during the summer term. Furthermore, I am involved in the UT Astronomy Vertically Integrated Projects (VIP; PI Finkelstein). VIP is a 3 year research program for undergrads led by graduates and post-docs, providing undergraduate STEM students with a multiyear research project, improving the retention of science majors at UT Austin. I plan to take on undergrad students after the creation of the sample detailed above and in preparation for data we will receive from *JWST* Early Release Science programs. Through these mentoring efforts, my goal is that, as a scientist, not only will I make meaningful contributions to the understanding of our Universe, but that throughout my career, I can help shift the dynamic of academia and research to one that supports scientists of all walks of life, particularly those less represented.

References ^[1] S. Tacchella, et al. 2019 ^[2] Stefanon et al., 2013 ^[4] Spitler et al., 2014 ^[5] Merlin et al., 2018 ^[6] Glazebrook et al. 2017 ^[7] Marchesini 2007 ^[8] Somerville et al. 2008 ^[9] Somerville et al. 2015